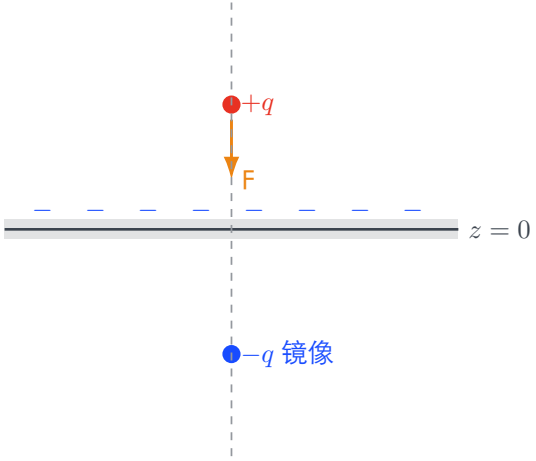
接地平面前的点电荷：物理半空间由等量异号镜像电荷构造



$$\varphi = kq\left[\frac{1}{\sqrt{s^2+(z-d)^2}} - \frac{1}{\sqrt{s^2+(z+d)^2}}\right]$$

$$\sigma(s) = -\frac{qd}{2\pi(s^2+d^2)^{3/2}}, \qquad Q_{\text{感}} = -q$$

$$\mathsf{F} = -\frac{kq^2}{4d^2}\mathsf{e}_z, \qquad U = -\frac{kq^2}{4d}$$